

Boolean algebra

Exercise 1

Provide the truth table for each of the following logic functions:

- $f(a, b, c) = ab + a\bar{b}c$
- $f(a, b, c) = \bar{a}b + bd + a\bar{c}$

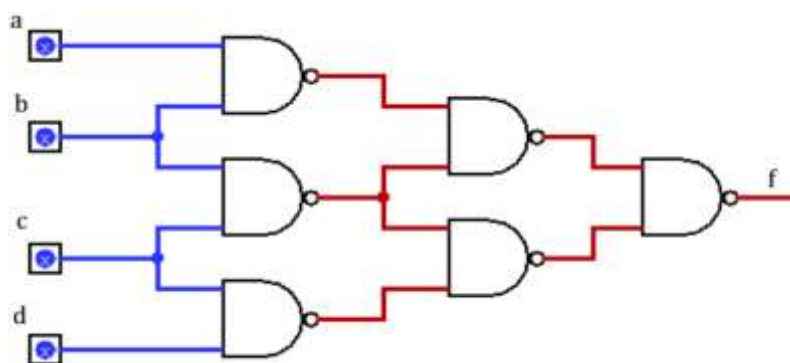
Exercise 2

Prove the following equalities using the Boolean algebra rules: `z

- $ab + bc + \bar{b}c = ab + c$
- $ab + \bar{a}c + bc = ab + \bar{a}c$
- $\bar{a} + ab + a\bar{c} + a\bar{b}\bar{c} = \bar{a} + b + \bar{c}$
- $(a + b)(\bar{a} + c)(b + c) = (a + b)(\bar{a} + c)$
- $\bar{a}d + b\bar{c}d + ab\bar{c} = \bar{a}d + ab\bar{c}$

Exercise 3

Consider the following logic circuit



1. Determine the function f of this circuit
2. Simplify f using De Morgan's theorem and draw the equivalent circuit of the function f

Exercise 4

Simplify the followings expressions by using Boolean algebra properties:

- $\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$
- $(A + B + C)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + \bar{C})$
- $\bar{A}B\bar{C} + \bar{A}BC + \bar{A}\bar{B}C + ABC$
- $(A + B + C)(\bar{A} + B + C)(\bar{A} + \bar{B} + C)(\bar{A} + B + \bar{C})$
- $B + A(B + C) + B(B + C)$
- $\overline{\bar{A}\bar{B}C + \bar{B}C + \bar{A}\bar{C}}$

Exercise 5

Let f and g two logic functions such as:

- $f(a, b, c) = abc + \bar{a}b + \bar{b}c$
- $g(a, b, c) = a\bar{b} + \bar{b}c$

1. Express F in canonical SoP form.
2. Express G in canonical PoS form.
3. Simplify f and write it only with NAND $\uparrow$.
4. Simplify g and write it only with NOR $\downarrow$.

Exercise 6

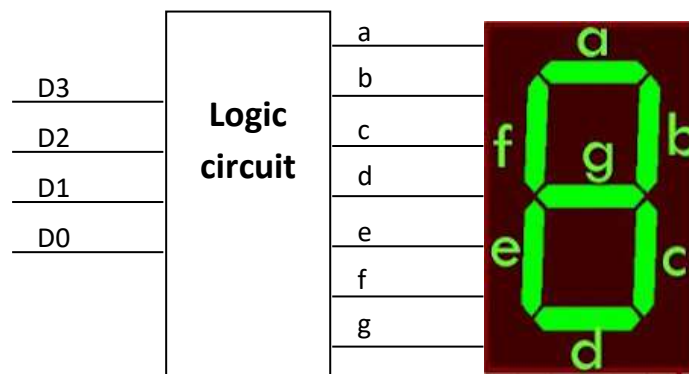
In a brick factory, quality control is carried out according to four criteria: **weight P, length L, height H and width W (0 incorrect, 1 correct)**. This makes it possible to classify bricks into three categories:

- **QUALITY A:** the weight P and at least two dimensions are correct.
- **QUALITY B:** the weight alone is incorrect or, the weight being correct, at least two dimensions are incorrect.
- **QUALITY C:** the weight P is incorrect and at least one dimension is also incorrect.

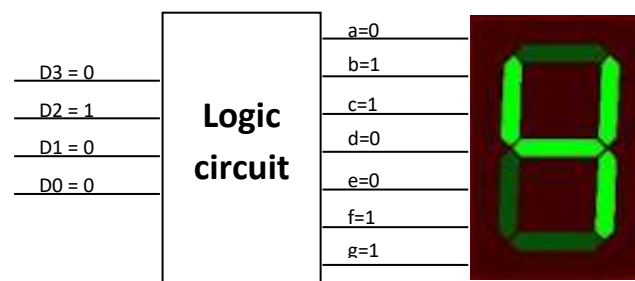
1. Write the functions A, B in SoP canonical form and C in PoS canonical form.
2. Simplify these functions using the Karnaugh Map method.
3. Draw the logic circuit diagram of outputs A, B and C.

Exercise 7

We want to create a digital display tool called a seven-segment display, the structure of which is shown below:



Such as D0, D1, D2 and D3 are the bits that make up the decimal digit to be displayed

Example of display number 4:

1. Give the truth table of this system
2. using the KARNAUGH map, give the simplified logic functions a, b, c, d, e, f and g
3. draw the functions diagram